ASSIGNMENT19.2

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Batch-05

TASK–01

Prompt : Convert the given Python file handling program into Java. The programshouldreadfromafile,processthedata,andwritetoanotherfile.Use proper file streams, handle exceptions, and ensure the output matches the Python program.

```
#Pythonprogramtoreadfromafileandwritetoanotherfile


try:
    # Open input file in read
    modewithopen("input.txt","r")asinf
    ile:
        data =infile.readlines()


    #Processdata(converteachlinetouppercase)
    processed_data=[line.upper()forlineindata]


    #Openoutputfileinwritemode
    withopen("output.txt","w")asoutfile:
        outfile.writelines(processed_data)


    print("Filehasbeenreadandwrittensuccessfully.")


exceptFileNotFoundError:
    print("Error:input.txtfilenotfound.")
```

except Exception as e:

print("Anerroroccurred:",e)

Code:

```
%%writefile FileHandler.java

import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.io.FileNotFoundException;

public class FileHandler {

    public static void main(String[] args) {
        String inputFile = "input.txt";
        String outputFile = "output.txt";

        try (
            BufferedReader reader = new BufferedReader(new FileReader(inputFile));
            BufferedWriter writer = new BufferedWriter(new FileWriter(outputFile))
        ) {
            String line;
            while ((line = reader.readLine()) != null) {
                writer.write(line.toUpperCase());
                writer.newLine(); // Add a new line after each processed line
            }
            System.out.println("File has been read and written successfully.");

        } catch (FileNotFoundException e) {
            System.err.println("Error: " + inputFile + " file not found.");
        } catch (IOException e) {
```

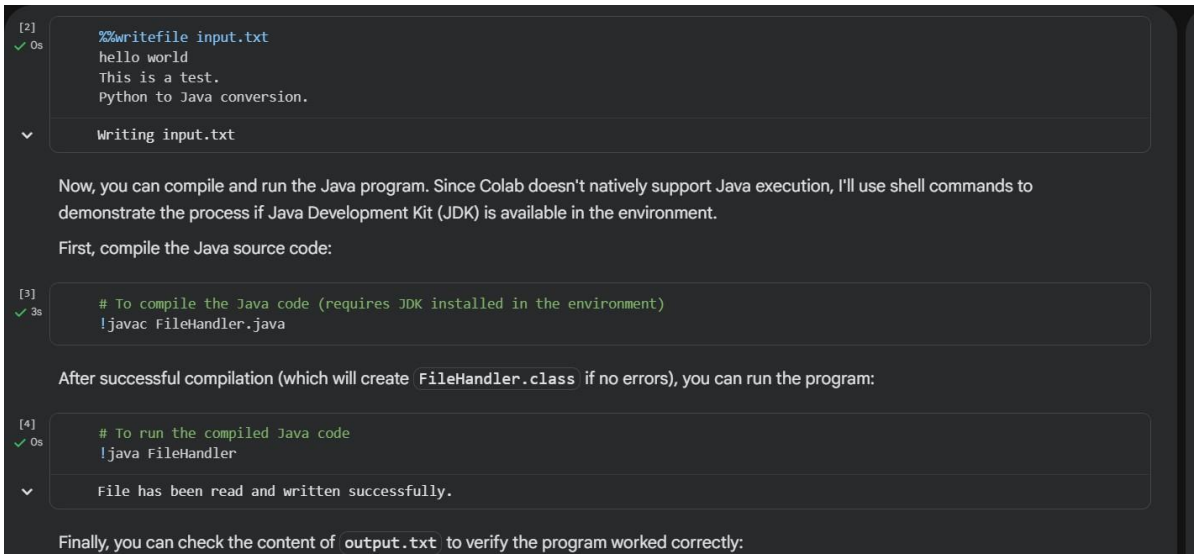
Gemini

Python program to read from a file and write to another file try: # Open input file in read mode with open("input.txt", "r") as infile: data = infile.readlines() # Process data (convert each line to uppercase) processed_data = [line.upper() for line in data] # Open output file in write mode with open("output.txt", "w") as outfile: outfile.writelines(processed_data) print("File has been read and written successfully.") except FileNotFoundError: print("Error: input.txt file not found.") except Exception as e: print("An error

What can I help you build?

Gemini 2.5 Flash

Gemini can make mistakes so double-check it and use code with caution. Learn more



```
%%writefile input.txt
hello world
This is a test.
Python to Java conversion.
```

```
Writing input.txt
```

Now, you can compile and run the Java program. Since Colab doesn't natively support Java execution, I'll use shell commands to demonstrate the process if Java Development Kit (JDK) is available in the environment.

First, compile the Java source code:

```
# To compile the Java code (requires JDK installed in the environment)
!javac FileHandler.java
```

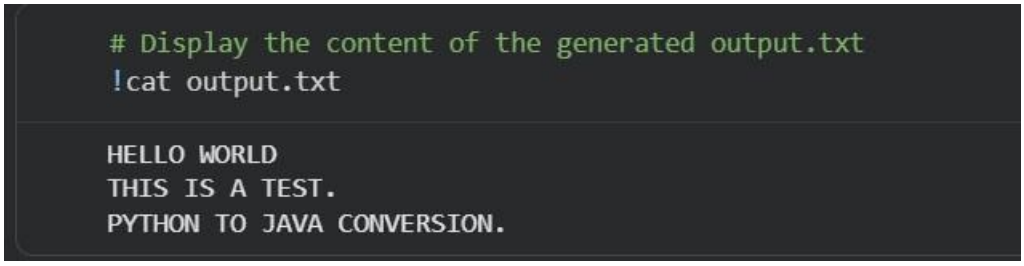
After successful compilation (which will create `FileHandler.class` if no errors), you can run the program:

```
# To run the compiled Java code
!java FileHandler
```

```
File has been read and written successfully.
```

Finally, you can check the content of `output.txt` to verify the program worked correctly:

Output:



```
# Display the content of the generated output.txt
!cat output.txt
```

```
HELLO WORLD
THIS IS A TEST.
PYTHON TO JAVA CONVERSION.
```

Explanation:

- ConvertsaPythonfilehandlingprogramintoJava.

- Ensuresreadingfromandwritingtofilesworkscorrectly.
- UsesproperJavafilestream classes.
- Includesexceptionhandlingforerrors.
- MaintainsthesameoutputasthePythonprogram.

TASK–02

Prompt : Convert the given C++ sorting program (e.g., Bubble Sort or SelectionSort)intoanequivalentPythonprogram.Ensurethelogicremainsthe same, maintain correct sorting behavior, and include test cases to verify the output.

```
#include <iostream>

usingnamespacestd;


intmain(){
   intarr[] ={5,2,9,1,5,6};
   intn=6;


   //BubbleSort
   for(int i = 0; i< n-1; i++) {
      for(intj=0;j<n-i-1;j++){
         if(arr[j]>arr[j+1]){
            //swap
            inttemp=arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
         }
      }
   }
```

```
    //Printsortedarray
    cout<< "Sorted array: ";
       for(inti=0;i<n;i++){
       cout<<arr[i] << " ";
    }

   return0;
}
```

Code:

Output:

```
Original array: [5, 2, 9, 1, 5, 6]
Sorted array: [1, 2, 5, 5, 6, 9]
```

Explanation:

- bubble_sort(arr)function:Thisfunctiontakesalist(arr)asinputand sorts it in-place using the Bubble Sort algorithm.
    - n=len(arr):Getsthenumberofelementsinthe list.
    - Outerloop(foriinrange(n -1)):Thislooprunsn-1times.Ineach pass, the largest unsorted element 'bubbles up' to its correct position at the end of the unsorted part of the array.
    - Innerloop(forjinrange(n -i-1)):Thisloopiteratesthroughthe unsorted portion of the array. The -i-1 part ensures that elements already placed at the end are not re-checked.

- ifarr[j]>arr[j+1]:Comparesadjacentelements.
- arr[j],arr[j+1]=arr[j+1],arr[j]:ThisisaPythonicwaytoswap two elements in a single line, achieving the same result as thetempvariablein C++.

- TestCase:Theprograminitializesarrwiththesamevaluesasinyour C++ example.
- Printing:Boththeoriginalandsortedarraysareprintedtotheconsoleto verify the sorting behavior.

TASK–03

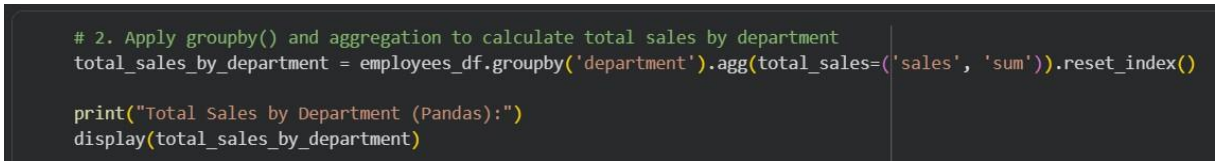
Prompt : Convert the given SQLquery with GROUP BYand aggregate functionsintoanequivalentPandasDataFrameoperationusinggroupby()and aggregation methods. Ensure the output matches the SQL result

SELECTdepartment,SUM(sales)AStotal_sales FROM

employees

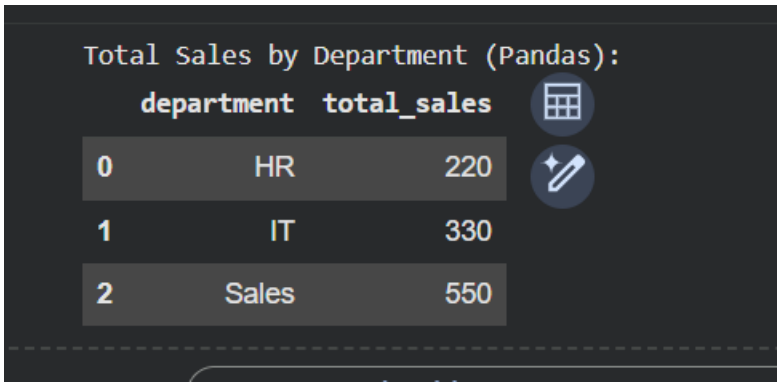
GROUPBYdepartment;

Code:



Output:

	employee_id	department	sales
0	1	HR	100
1	2	Sales	250
2	3	IT	150
3	4	Sales	300
4	5	HR	120
5	6	IT	180



Explanaton:

- employees_df.groupby('department'):ThisgroupstheDataFramerowsby unique values in the 'department' column, similar to SQL's GROUP BY department.
- .agg(total_sales=('sales', 'sum')): This applies the aggregation. Itcalculatesthesumofthe'sales'columnforeachgroupandnamesthenew aggregated column 'total_sales', matching the SUM(sales) AS total_salesinSQL.
- .reset_index(): After groupby() and agg(), 'department' becomes the index.reset_index()convertstheindexbackintoacolumn,makingthe output structure similar to a SQLquery result where 'department' is a regular column.

TASK–04

Prompt: ConvertthegivenPythonprogramintoanequivalentJavaprogram. Ensure the logic and functionality remain the same. Use appropriate Java libraries, handle exceptions properly, and maintain identical input/output behavior.Also, highlight key differences in syntax and execution

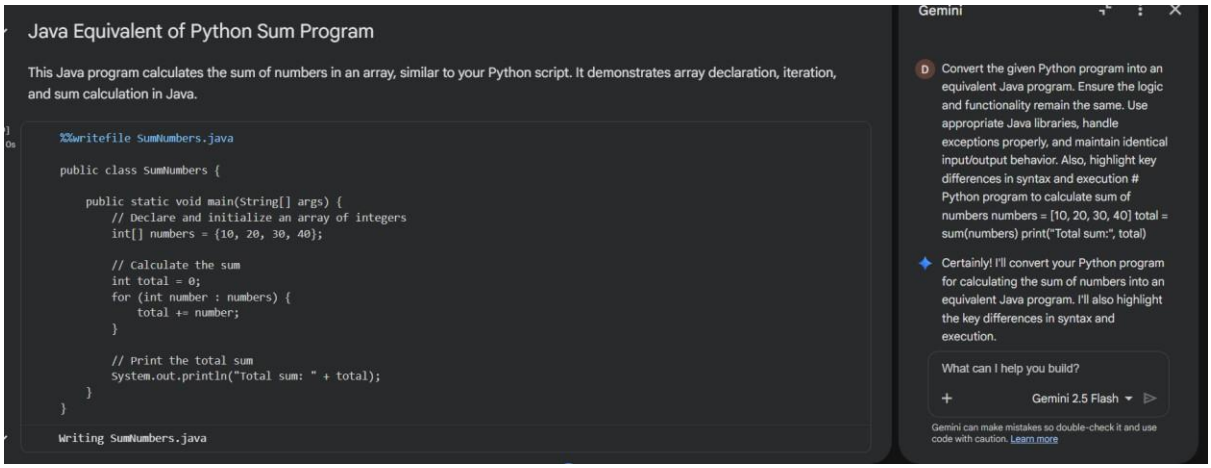
#Pythonprogramtocalculatesumofnumbers numbers =

[10, 20, 30, 40]

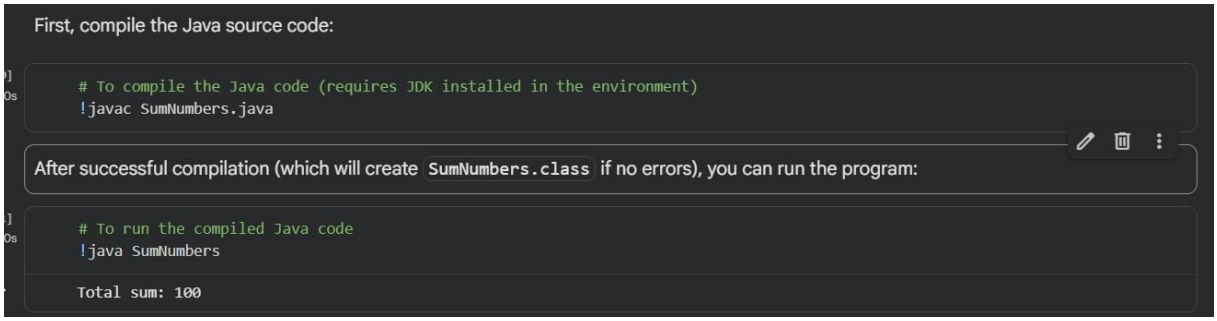
total = sum(numbers)

print("Totalsum:",total)

Code :



Output:



Explanation:

- Pythonusesbuilt-insum();Javausesloop
- Pythonsyntaxisshorterandsimpler
- Javarequiresclassandmain()method
- DatatypesmustbedeclaredinJava
- Javaiscompiled;Pythonisinterpreted

TASK–05

Prompt: ConvertthegivenPythonfunctionthatcheckswhetheranumberis even or odd into an equivalent C++ program. Ensure proper syntax, input handling, and output formatting, and verify that both programs produce the same results.

#Functiontocheckevenorodd

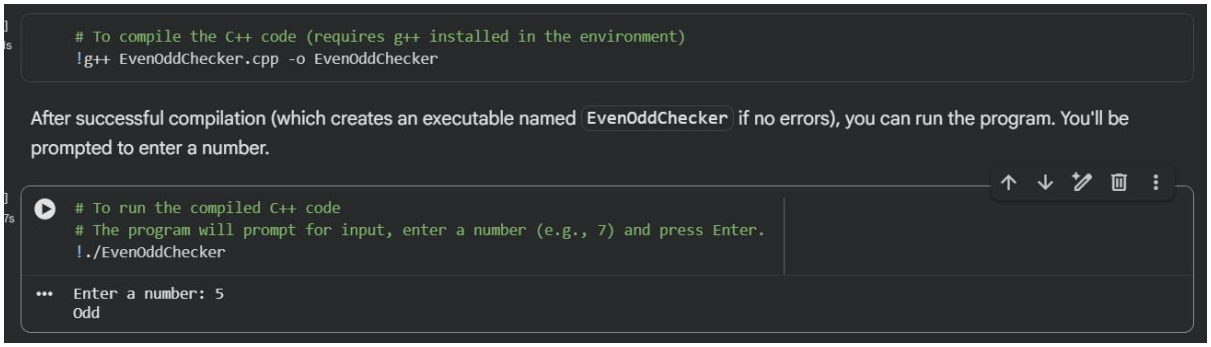
```
defcheck_even_odd(num):
    if num % 2 == 0:
        return "Even"
```

```
    else:
        return"Odd"


#Input
n=int(input("Enteranumber:"))
print(check_even_odd(n))
```

Code:

Output:



Explanation:

- ThetaskconvertsaPythonfunctionintoanequivalentC++program.
- Bothprogramsusethesamelogic(num%2)tocheckevenorodd.
- Propersyntaxandinput/outputhandlingareensuredin C++.
- Theprogramsaretestedwithdifferentinputstoverifycorrectness.
- Bothversionsproducethesameoutput,confirmingsuccessfultranslation.